

SRG-I Mathematical source reconstruction

Symbolic Resonance Geometry

Historical Reconstruction, Recursive Well-Posedness, and Dual-Sector Symmetries

Hilmir Frímann Halldórsson

Revision date: 10 October 2026

Revised manuscript, v0.1.1, returned for closure review. Date above is not a publication date.

Abstract

The surviving Symbolic Resonance Geometry sources define several state schemas, operator families and sector relations. We reconstruct their mathematical content without selecting a replacement evolution. Literal substitution of a July 2025 temporal second difference into its accompanying update reduces the scalar problem to $y = C_\lambda(y + b)$. For the printed golden-ratio threshold scaling we prove the complete solution set for every real threshold. At positive threshold, some offsets have no successor, zero offset has a continuum of successors, and a restricted positive branch has exactly one. The result addresses algebraic existence and uniqueness for fixed current data, not full well-posedness of the symbolic state. We distinguish compression maps by carrier, sign, zero and branch behavior, and separate temporal differences, spatial graph coupling and alternative toroidal metrics. The June phase pair defines a quarter-turn and real quadrature; July time shift, geometric reflection and later sector exchange require different domains and symmetry conditions. Prescribed clock modulation, trail embeddings and a later driven memory equation likewise encode different kinds of memory. Proofs and counterexamples identify recoverable structure while leaving missing state updates, fusion rules, operator composition and physical interpretations explicit. Later models serve as dated comparisons, not retrospective completions.

1 Introduction: reconstruction before completion

A recursion must specify more than a recurring pattern or a relationship between quantities. To obtain a next-state map, the state carrier, required history, external inputs, and update operations must be defined sufficiently to determine a successor on a stated domain. An implicit relation may instead have no successor or several. An observer can describe a completed trajectory without generating one. These distinctions guide the reconstruction below.

The historical corpus uses shared names for constructions that are not the same mathematical object. The formation manuscript P03 describes persistent glyph identities, scalar resonance, positions, memory, and echo sets; it also prints more than one expression for resonance evolution. The numerical operator manuscript P07 and the later reconstruction P05 use additional, distinct operator definitions. The June phase constructions P09 and P10 supply a relation between two phase-labelled fields. Their coexistence does not by itself establish a single composed dynamics [P03, physical pp. 2–8] [P07, physical pp. 3–6] [P05, physical pp. 1–3] [P09, physical p. 2] [P10, physical p. 2].

The question addressed here is therefore not which missing rule should now be invented, but what the surviving definitions already determine. The principal result is an exact scalar branch classification for the implicit relation obtained from P03. The distinction between supported phase identities and unspecified dynamical symmetries supplies a second part of the analysis.

The present text follows the accepted source reconstruction and branch appendix, with proof details included rather than replaced by numerical examples [F01, B].

1.1 Source and evidence conventions

Identifiers P01–P19 are stable identifiers within the retained archive, not publication identifiers. Citations specify physical PDF pages, counted from one. The formation paper P03 carries a cover date of 14 July 2025; P07 carries 11 July 2025; P09 and P10 carry 29 June 2025. A cover date, a PDF metadata date, and a date on which a file was copied are different kinds of evidence. The source catalog retains that distinction [C].

We separate three roles throughout. A *historical definition* records what a manuscript or program states. A *deduction* follows from explicitly stated mathematical assumptions. A *completion choice* supplies additional structure that the bounded source chain does not determine. A completion choice can be mathematically useful without being recovered historical intent. In particular, a scalar interpretation of a symbolic tuple must be declared, not inferred from notation alone.

The existing Tower II mechanism studies concern a later, specified executable model. Their completed trajectories do not determine the meaning of an incomplete 2025 formula. This paper is not a new simulation study and does not claim that every historical SRG implementation is described by the relation examined here [F01].

1.2 Chronology and limits of the comparison

The central source chain comprises the June modules and tensor memo, the 11 July numerical operators, the 13 July signal-sector discussion, the 14 July formation paper and the 15 July fusion/reconstruction papers. P11 and P13 have 29 June PDF creation metadata but no established cover date; this is not evidence of publication priority. The November geometric and finite-core manuscripts and the March 2026 real coupled-field program are dated comparisons. Their carriers and operators are not inserted into the July recursion [P01, P05, P06, P08, P11, P13, P14, P15, Infinity]. The working title uses *Symbolic Resonance Geometry*; the bibliography retains each source’s actual title rather than asserting a uniform historical expansion of SRG.

Elementary identities, differentiation, term tests and the contraction principle are standard tools. The contribution here is their source-specific application, particularly the exact successor classification and the separation of incompatible definitions. Priority beyond the bounded corpus has not been assessed.

2 State schemas and distinct historical formulas

2.1 A glyph schema is not yet a closed state space

P03 defines a glyph by the schema

$$Q_i = (\text{ID}_i, R_i(t), \mathbf{X}_i(t), H_i(t), \mathcal{E}_i), \quad (2.1)$$

Here glyph subscripts and the short label ID_i are presentational renamings of the source tuple, not extra state laws. The entries describe persistent identity, resonance, position, memory, and an echo set. The same manuscript later uses Θ_i for a symbolic state that retains resonance, trail direction, and identity [P03, physical pp. 2,18].

Equation (2.1) is a useful schema. It does not declare an algebra in which a scalar can be added to an ID or a set, a norm of the complete tuple, or updates for all its components. A scalar

compressor can act on a selected numeric component; applying it to the entire tuple requires additional structure. Consequently, identifying Θ_i with R_i is a modeling interpretation, not an equality established by the printed schema.

2.2 An explicit expression, a consistency expression, and an implicit one

Three source formulas must be kept distinct. P03 first writes

$$R_i(t + \Delta t) = C \left[R_i(t) + \sum_{j \in \mathcal{E}_i} f(R_j(t), d_{ij}, H_i(t)) \right] \quad (2.2)$$

and elsewhere gives

$$R_i(t) = H_i(t) \sum_{j \in \mathcal{E}_i} f(R_j(t), d_{ij}). \quad (2.3)$$

Its trail-based update is

$$\Theta_i(t + 1) = C \left[\Theta_i(t) + \nabla_T^2 \Theta_i(t) + F_i(t) \right]. \quad (2.4)$$

These appear respectively on physical pages 2, 5, and 7 [P03, §§1.1,2.2,3.1].

Equation (2.2) is first order once its right-hand data and compressor are supplied. Equation (2.3) is an instantaneous consistency expression. For example, with an empty echo set it forces zero resonance, while (2.2) can preserve a nonzero value in an identity branch of the compressor. Thus they cannot both be imposed on arbitrary initial amplitudes without a further interpretation. Selecting the explicit expression as the intended replacement for (2.4) would be an editorial decision, not a consequence of the shared notation [F01, §§2.1,3.4].

We write F_i for the sum of the source's individual feedback terms; this is shorthand, not an additional force law. Using its printed summand, the feedback in (2.4) is

$$F_i(t) = H_i(t) \sum_{j \in \mathcal{E}_i(t)} \frac{R_j(t) \cos \varphi_{ij}(t)}{1 + d_{ij}(t)^2}. \quad (2.5)$$

It becomes a scalar expression when its arguments are specified. A complete model must still state the echo-set selection/update, the distance convention, the phase or angular-alignment rule, and the memory input. A dimensional distance also requires a scale in the denominator; choosing a dimensionless distance is one possible convention, not a derived physical calibration [P03, physical p. 7, §3.2] [F01, §2.2].

Reciprocal echo membership identifies a two-way graph connection. It does not, without a specified evolution and a state-return condition, prove a periodic orbit. Similarly, a predicate that two glyphs are close and have similar resonance does not specify the amplitude, ID, memory, or echo set of a merged glyph [P03, physical pp. 3,10,12] [F01, §§2.3–2.4].

2.3 Feedback reciprocity, fusion and initialization

The source feedback has useful conditional structure. If distance is symmetric, $\varphi_{ji} = -\varphi_{ij}$, and echo membership is reciprocal, then $a_{ij} = \cos \varphi_{ij} / (1 + d_{ij}^2)$ is symmetric on the active edges. The actual coefficient in (2.5) is $H_i a_{ij}$. Under the stated symmetry of a_{ij} , equality of the two directional coefficients is equivalent to $(H_i - H_j) a_{ij} = 0$. Thus the memory values must agree on nonzero-weight edges; a zero-weight edge imposes no such condition. Even when the coefficients are symmetric, there is no compensating term proportional to $-R_i$. For a mutually coupled pair with positive resonances, aligned phases and positive memory, both contributions

are positive. Symmetric coupling therefore does not establish total-resonance conservation. This is a deduction about the printed feedback, not a conservation test on an executed trajectory [P03, physical p. 7, §3.2].

The corridor criteria include

$$A \in \mathcal{E}_B, \quad B \in \mathcal{E}_A, \quad |R_A - R_B| < \epsilon_R, \quad d_{AB} < \lambda_{VP}, \quad |\varphi_{AB}| < \theta_{lock}. \quad (2.6)$$

The nearby merge statement has the form

$$d_{AB} < \lambda_{VP}, \quad R_A \approx R_B \quad \implies \quad Q_{AB}^{\text{fused}} = \text{ProtoSRG} \quad (2.7)$$

[P03, physical p. 10, §§4.1–4.3]. This records a proposed recognition condition. The right side supplies a name, but not the entries of the tuple (2.1), an ID assignment, a deletion rule for the parents, or an update order. Identifying the approximation in (2.7) with the ϵ_R tolerance is plausible but still a declared convention. The cluster condition on physical p. 12 similarly specifies proximity and amplitude thresholds; it does not define an invariant set with a basin of attraction.

The fusion paper instead describes an observer pipeline: resonance peaks are normalized, events $(X, Y, Z, R_{\text{norm}})$ are clustered, and events with the same ID are linked in pass order. Its clustering radius is related to λ_{VP} , but the relative scale of spatial coordinates and normalized amplitude in that distance remains a choice [P01, physical pp. 5–7, §§3.2–3.5]. The narrative that several trails reach a common glyph identity is not an algorithm for merging different IDs. An observer applied to recorded events and a rule generating those events have different mathematical roles.

P03 calls a pass a computational index rather than continuous physical time [P03, physical p. 7, §3.1]. The centered temporal expression below requires both a current and a previous numeric state. A seed at a single pass does not provide the second time level. No convention for that initialization, nor for the direction of a stationary trail used in phase alignment, is recovered. These are explicit domain questions, independent of whether a later scalar branch equation happens to have a root.

3 The implicit scalar relation and its complete branches

3.1 Literal reduction

P03 defines its trail Laplacian by the centered temporal difference

$$\nabla_T^2 \Theta_i(t) = \Theta_i(t+1) - 2\Theta_i(t) + \Theta_i(t-1). \quad (3.1)$$

No additional coefficient or time-step denominator is printed in this definition or before the term in (2.4) [P03, physical p. 8, §3.3].

For the following theorem, interpret the evolving quantity as a finite real scalar and freeze the current-pass feedback data. Set

$$y = \Theta_i(t+1), \quad b = \Theta_i(t-1) - \Theta_i(t) + F_i(t). \quad (3.2)$$

Substitution, without changing any source coefficient, gives

$$\boxed{y = C_\lambda(y + b)}. \quad (3.3)$$

The appearance of the future state on both sides is not by itself an error: an implicit equation can define a valid update. The question is whether this particular relation has a root and whether that root is unique.

3.2 Threshold definition and classification

Write $\phi = (1 + \sqrt{5})/2$ and $q = \phi^{-1}$. To distinguish the threshold from other historical parameters, denote it by λ throughout this section. P03's printed threshold compressor is

$$C_\lambda(x) = \begin{cases} x, & x < \lambda, \\ qx, & x \geq \lambda. \end{cases} \quad (3.4)$$

Equality belongs to the compressed branch [P03, physical p. 8, §3.4].

Theorem 3.1 (Complete real scalar solution set). *Let $b, \lambda \in \mathbb{R}$ and let C_λ be defined by (3.4). Define*

$$\mathcal{L}(b, \lambda) = \begin{cases} (-\infty, \lambda), & b = 0, \\ \emptyset, & b \neq 0, \end{cases} \quad (3.5)$$

$$\mathcal{U}(b, \lambda) = \begin{cases} \{\phi b\}, & \phi^2 b \geq \lambda, \\ \emptyset, & \phi^2 b < \lambda. \end{cases} \quad (3.6)$$

Then the complete solution set of (3.3) is

$$\mathcal{S}(b, \lambda) = \mathcal{L}(b, \lambda) \cup \mathcal{U}(b, \lambda). \quad (3.7)$$

Proof. The input $y + b$ lies in exactly one of the two source branches. If $y + b < \lambda$, the equation is $y = y + b$. It requires $b = 0$, and then every $y < \lambda$ satisfies both the equation and its branch condition.

If $y + b \geq \lambda$, then $(1 - q)y = qb$. The golden-ratio identity $\phi^2 = \phi + 1$ gives $q/(1 - q) = \phi$, so the only candidate is $y = \phi b$. Its input is $(\phi + 1)b = \phi^2 b$; hence this candidate is admissible exactly when $\phi^2 b \geq \lambda$. Both branches have been exhausted, and each listed value satisfies its own branch condition. This establishes necessity and sufficiency. $\square$

Corollary 3.2 (Positive threshold). *For $\lambda > 0$, the number and form of real scalar successors are as follows.*

Offset	Complete real solution set
$b < 0$	$\emptyset$
$b = 0$	$(-\infty, \lambda)$
$0 < b < \lambda/\phi^2$	$\emptyset$
$b \geq \lambda/\phi^2$	$\{\phi b\}$

Proof. For a positive threshold, the upper-branch condition forces $b \geq \lambda/\phi^2 > 0$. The lower branch exists only at $b = 0$. Equation (3.7) therefore yields the table. $\square$

At the boundary $b = \lambda/\phi^2$, the successor is $y = \lambda/\phi$ and its input equals λ , as required by the source's upper-branch convention. If nonnegative amplitudes are required, intersect the solution set with $[0, \infty)$. For $\lambda > 0$ this changes the $b = 0$ continuum to $[0, \lambda)$ and leaves the other rows unchanged. At $\lambda = 0, b = 0$, the real solution set is $(-\infty, 0]$, whose nonnegative part is $\{0\}$. Theorem 3.1 also covers negative thresholds without extending the positive-threshold table outside its domain [F01, §3.2].

3.3 Examples, typing, and what the theorem does not say

Example 3.3 (Failure of universal existence and uniqueness). Take $\lambda = 1$ and set the previous and current scalar states to zero. Then $b = F$. For $F = 0$, both $y = 0.2$ and $y = 0.7$ are roots, and the entire interval $(-\infty, 1)$ consists of roots. For $F = 0.1$, the uncompressed branch requires $b = 0$, while the compressed candidate has input $0.1\phi^2 \approx 0.2618 < 1$ and is inadmissible. There is no root. For $F = 1$, the unique root is $y = \phi$.

Thus the printed relation is not a total, single-valued scalar update over all finite inputs. Zero feedback defeats universal uniqueness; a small positive feedback can defeat existence even under a nonnegative-state restriction. A choice of initial guess or solver tolerance cannot supply a root when the exact solution set is empty. Conversely, choosing one member of a continuum of roots requires a stated selection rule.

For finitely many scalar components with all current-pass feedback fixed, the solution set is the Cartesian product of the componentwise sets. Existence requires that every factor be nonempty, and uniqueness requires that every factor be a singleton. If feedback, phase, or echo membership depends on the unknown successor, this product classification does not solve the newly coupled problem. Nor does a componentwise interpretation define an algebra on the tuple (2.1) [F01, §3.2].

These conclusions concern algebraic existence and uniqueness. They do not establish continuous dependence, a complete multistep trajectory, stability, or an attractor. The bounded source investigation did not identify an implicit solver or branch selector for this relation. The fusion paper’s listing is expressly pseudocode: its resonance-update stub copies and returns the input, and the text acknowledges the placeholders [P01, physical pp. 22–24]. The formation manuscript P03 also contains the explicit expression (2.2) [P03, physical p. 2, §1.1]; choosing it as a correction to (3.3) remains a separate decision [F01, §3.4].

3.4 Why the word compression does not settle solvability

For $\lambda > 0$, the one-sided limits of (3.4) at the threshold are λ and $q\lambda$, so the map has a downward jump. It is not a globally Lipschitz contraction. Its range is also unbounded above, because $C_\lambda(x) = qx \rightarrow \infty$. Finally, every $z \in [q\lambda, \lambda)$ has two distinct preimages, z and z/q . These properties follow from the specific printed map; they are not claims about every operator bearing the name RPCO [F01, §3.3].

4 Compression and operator identity

The earlier partner-dependent expression in P03 is $C(x) = \min(x, qR_{\text{partner}})$. With a fixed partner value and cap m , its substitution into the same implicit structure gives

$$y = \min(y + b, m), \quad \mathcal{S}_{\min}(b) = \begin{cases} \emptyset, & b < 0, \\ (-\infty, m], & b = 0, \\ \{m\}, & b > 0. \end{cases} \quad (4.1)$$

Indeed $y < m$ requires $b = 0$, whereas $y = m$ requires $b \geq 0$. This is a different branch classification. If the partner is itself an unknown next-state value, the fixed-cap argument no longer settles the coupled system [P03, physical p. 6, §2.5] [B, §1].

For $L > 0$ and $\epsilon > 0$, the numerical-operator manuscript P07 supplies another map:

$$C_+(x) = \exp(\text{clip}(\text{sgn}(x) \log(|x| + \epsilon), -L, L)). \quad (4.2)$$

Here the sign is *inside* the exponential. The later signed map places it outside:

$$C_s(x) = \text{sgn}(x) \exp(\text{clip}(\log(|x| + \epsilon), -L, L)). \quad (4.3)$$

The literal expressions imply $C_+(0) = 1$ and $C_s(0) = 0$. For negative inputs, the first has positive output while the second preserves the negative sign. The accepted source comparison records that P07's formula and its listing agree, despite adjacent sign-preservation prose; changing the formula to match the prose would be a correction rather than transcription [P07, physical p. 3] [F01, §4.1].

Neither logarithmic map should be substituted for (3.4) without declaring a different model. Such substitution also does not automatically restore unique implicit solvability: for the later defaults, $y = C_s(y)$ has precisely the three roots $0, \pm e^L$. Appendix A supplies the exhaustive alternate branch candidates and qualifications [B, §§2–3].

4.1 Exact sign and zero behavior

Use $a_- = e^{-L}$ and $a_+ = e^L$ for the lower and upper plateaus, and define $\text{clip}(u, a_-, a_+) = \min(a_+, \max(a_-, u))$. The positive-output expression (4.2) is exactly

$$C_+(x) = \begin{cases} \text{clip}(x + \epsilon, a_-, a_+), & x > 0, \\ 1, & x = 0, \\ \text{clip}((|x| + \epsilon)^{-1}, a_-, a_+), & x < 0. \end{cases} \quad (4.4)$$

This follows by monotonicity of the exponential and the identity $e^{-\log u} = 1/u$. At the printed defaults $L = 10$ and $\epsilon = 10^{-6}$, for instance, $C_+(-2) = (2 + \epsilon)^{-1} > 0$. The adjacent prose says that sign is retained, while the equation and code both give (4.4) [P07, physical p. 3, §2.1]. Neither version may be silently substituted for the other on the basis of that prose.

For the later program's $L = 9$ and $\epsilon = 10^{-9}$,

$$C_s(x) = \begin{cases} \text{sgn}(x) \text{clip}(|x| + \epsilon, a_-, a_+), & x \neq 0, \\ 0, & x = 0. \end{cases} \quad (4.5)$$

The definition is odd and preserves nonzero signs, but is discontinuous at zero because the one-sided limits have magnitude $\max(a_-, \epsilon) > 0$. Its plateaus lose magnitude information [Infinity, lines 49–53]. At $b = 0$, the relation $y = C_s(y + b)$ has exactly the three roots $0, \pm a_+$ for these defaults. Appendix A proves this and lists all sign and plateau candidates without choosing an operator for the original implicit recurrence.

4.2 A name does not fix an operator or its carrier

Other surviving constructions make the distinction especially clear. The original spatial RPCO helper samples

$$\rho_{\text{RPCO}}(x, y) = \sin(5 \text{atan2}(y, x)) e^{-0.05(x^2 + y^2)}. \quad (4.6)$$

It generates a field value from position; it is not an update of an incoming resonance [RPCO, lines 5–13]. The velocity-based program instead applies componentwise damping

$$D_v(v) = \frac{v}{1 + \log(1 + |v|)}. \quad (4.7)$$

For $v > 0$ its derivative is

$$\frac{1 + \log(1 + v) - v/(1 + v)}{(1 + \log(1 + v))^2} > 0.$$

The map is odd, strictly increasing, reduces the magnitude of each nonzero input, and is unbounded as $|v| \rightarrow \infty$. Thus damping need not be a cap [Attractor, lines 29–31]. A separate early vector-field program leaves vectors below magnitude 5 unchanged and multiplies the others by 0.2; that threshold filter is also unbounded [Helicity, lines 28–32].

The November finite-core model uses a linear partial projection $(1 - \lambda_c)I + \lambda_c P_{\text{core}}$ on a complex vector space [P15, physical p. 5, §3.1]. It neither thresholds a real scalar nor evaluates (4.6). The neutral organizing principle is the carrier, domain and literal map. RPCO, TGMO and REFU remain historical names, but their reuse supplies no equality of operators.

Source family	Input and operation	Information needed for use
P03 partner cap	Scalar, partner-dependent minimum	Partner identity and timing
P03 threshold	Scalar, identity/scaled branches	Threshold and admissible data
P07 logarithmic	Real input, positive output	Regularizer, clamp and composition
Original helper	Position to sampled field	Coordinate domain and sampling
Velocity program	Velocity to damped velocity	Force and time-step convention
Later signed code	Real amplitude to signed clamp	Stage order and parameters
P15 finite core	Complex vector, partial projection	Basis, projector and factor order

This table compares definitions; it is not a lineage asserting that one operation corrected or completed another.

5 Temporal curvature, spatial graphs and toroidal geometry

5.1 The index differentiated is part of the definition

The centered difference (3.1) acts along pass index and annihilates affine temporal sequences. A graph difference acts across simultaneous positions. They may both encode a form of variation, but their domains and required data differ. The source notation ∇_T^2 does not erase that distinction.

The reconstruction patch gives an explicit spatial engine [P05, physical pp. 1–2, §§2–3]. In its surviving code, positions form an array $X \in \mathbb{R}^{N \times 3}$, with one row per glyph. Let $P \in \mathbb{R}^{N \times N}$ average the initially selected neighbors on each nonempty row and use the identity row for an empty neighbor set. Write $\Delta X = (P - I)X$. The operative stages are

$$Y = X + 0.05\Delta X + 0.15 \tanh(\Delta X), \quad (5.1)$$

$$Z = 0.95Y + 0.05\bar{Y} + \xi, \quad (5.2)$$

$$X^+ = Z \operatorname{diag}(1, 1, 1 - 0.001 \sin(2\pi 0.244 s)). \quad (5.3)$$

The mean is broadcast to each position, the hyperbolic tangent acts componentwise, and ξ denotes the program’s Gaussian perturbation. In (5.3), right multiplication scales coordinate columns, so only each glyph’s third coordinate changes in that stage. The initial graph is built once. The variables named GAMMA, LAMBDA_VP and a computed breathing amplitude do not enter these operative updates [Position, lines 31–40,45–78]. Equations (5.1)–(5.3) are a static transcription of that program, not an execution here.

For a row-stochastic P , $(P - I)\mathbf{1} = 0$; constant positions have zero graph difference. On the other hand,

$$\mathbf{1}^T(P - I)X = 0 \quad \text{for every } X \quad \Longleftrightarrow \quad \mathbf{1}^T P = \mathbf{1}^T.$$

Thus unweighted-sum conservation additionally needs column balance. A directed nearest-neighbor graph need not have it, and subsequent nonlinear or forcing stages require their own

analysis. This conditional identity does not turn the graph operator into the printed temporal one.

The patch carries positions but not the complete resonance/echo schema (2.1). It supplies a surviving explicit alternative, not a source-identified solver for (3.3). Choosing temporal curvature as an observer, a spatial term, or a second-order state equation remains undecided by their coexistence.

5.2 What the Laplacian failure analysis establishes

The June failure note distinguishes a constant radius from a spatial radius field [P11, physical p. 1, §§1–3]. A constant has zero spatial derivatives; promoting it to $r(x, y) = \sqrt{x^2 + y^2}$ changes its type and its mathematical behavior. Off the origin,

$$r_{xx} = \frac{y^2}{r^3}, \quad r_{yy} = \frac{x^2}{r^3}, \quad \Delta r = \frac{1}{r}. \quad (5.4)$$

These identities follow by differentiation. At the origin r is not classically differentiable, so they do not supply a smooth global repair. The note’s reported array-gradient failure is a historical account; no failure case or old program was rerun for this paper.

There are also distinct spatial formulas. P08’s code differentiates the array named theta twice along two axes, with coefficients a^{-2} and r^{-2} , while its phi argument is unused. It supplies no spacing or periodic endpoint convention [P08, physical p. 2, §2.3]. P07 prints

$$\mathcal{L}f = a^{-2}\nabla_\theta^2 f + b^{-2}\nabla_\varphi^2 f + \nabla_r^2 f, \quad (5.5)$$

but its listing applies a function named laplacian to theta, phi and r without a field argument or a definition of that function [P07, physical p. 4, §2.2]. Differentiating coordinate arrays and differentiating a field with respect to coordinates are not the same operation. Calling an array gradient twice also does not impose toroidal boundary conditions.

5.3 A conditional embedded-torus diagnostic

The following calculation specifies its geometry; it is not a recovered replacement. For an embedded ring torus with major radius $R > r > 0$, let

$$\mathbf{x}(u, v) = ((R + r \cos u) \cos v, (R + r \cos u) \sin v, r \sin u). \quad (5.6)$$

The coordinate tangents are orthogonal with squared lengths r^2 and $(R + r \cos u)^2$. Hence $g = \text{diag}(r^2, (R + r \cos u)^2)$. For a smooth scalar field the standard Laplace–Beltrami formula gives

$$\begin{aligned} \Delta_g f &= \frac{1}{\sqrt{\det g}} \partial_a \left(\sqrt{\det g} g^{ab} \partial_b f \right) \\ &= \frac{f_{uu}}{r^2} - \frac{\sin u}{r(R + r \cos u)} f_u + \frac{f_{vv}}{(R + r \cos u)^2}. \end{aligned} \quad (5.7)$$

The first-order term follows by differentiating the volume factor $r(R + r \cos u)$; it cannot be discarded while retaining this induced metric.

For $f(u, v) = \cos u$ at $u = \pi/2$, $\Delta_g f = 1/(rR)$, whereas a flat constant-coefficient two-angle second-derivative operator gives zero there. A flat product torus can legitimately have the latter type of operator, but it is a different metric. The comparison distinguishes possible geometries without choosing between them.

The original TGMO helper samples

$$((2 + r \cos \theta) \cos(\pi/4), (2 + r \cos \theta) \sin(\pi/4), r \sin \theta) \quad (5.8)$$

[TGMO, lines 5–14]. Its fixed azimuth traces one minor circle, repeated by the parameter range. It specifies neither a full surface field nor a Laplacian. Wrapped coordinates, a torus-shaped drawing, a flat product metric and the induced metric of (5.6) therefore cannot be identified from nomenclature alone.

6 Dual-sector relations and meanings of reversal

6.1 The June complex phase pair

P09 and P10 define

$$\Phi_Q = e^{i\theta}T, \quad \Phi_D = e^{i(\theta+\pi/2)}T = i\Phi_Q. \quad (6.1)$$

This fixes a phase relationship independently of whether a complete update for T and θ has been specified [P09, physical p. 2, §2.3] [P10, physical p. 2, §2.3].

Proposition 6.1 (Order and orthogonality of the phase relation). *Let z belong to a complex vector space and define $Jz = iz$. Then $J^2 = -I$ and $J^4 = I$. If the space has a Hermitian inner product linear in its second argument, then*

$$\langle z, Jz \rangle = i\|z\|^2. \quad (6.2)$$

Thus a nonzero pair (z, Jz) is not orthogonal over the complex Hilbert space. It is orthogonal in the underlying real space equipped with $(u, v)_{\mathbb{R}} = \text{Re}\langle u, v \rangle$.

Proof. Successive multiplication by i gives $i^2 = -1$ and $i^4 = 1$. Linearity in the second inner-product argument gives the displayed identity, whose real part is zero. $\square$

The resulting sequence

$$z \mapsto iz \mapsto -z \mapsto -iz \mapsto z \quad (6.3)$$

is an algebraic orbit of J . It is not evidence that the source dynamics moves through these four values in consecutive time steps. Likewise, if one separately adopts a projective-state convention that quotients common phase, z and iz describe the same ray. The source relation does not supply a measurement theory identifying distinct particles [F01, §6.1].

Plain exchange of the constrained pair is different. Writing $E(A, B) = (B, A)$ gives $E^2 = I$, but $E(z, iz) = (iz, z)$ is not of the form (w, iw) unless $z = 0$. Advancing the common phase by $\pi/2$ instead produces $(iz, -z)$, a signed exchange. Complex conjugation, spatial reflection, and reversal of a specified evolution are further distinct operations. Their use as a common word such as “mirror” cannot replace a declaration of their actions and domains [F01, §6.1] [B, §6].

The July relation $D(t) = Q(t + \tau)$, with $\tau = \pi/(2\omega)$, gives multiplication by i for the specified mode $Q(t) = e^{i\omega t}$. It does not do so for an arbitrary signal. A constant is unchanged, while a 2ω mode acquires a phase of π . A time shift therefore requires its own function-space and sampling conventions; it is not automatically a time-reversal operation [P06, physical pp. 3,8] [F01, §6.3].

6.2 Time shift, reflection and reversal

For functions on a two-sided time domain, define $(S_\tau f)(t) = f(t + \tau)$ and $(Tf)(t) = f(-t)$. Then

$$S_\tau^{-1} = S_{-\tau}, \quad T^2 = I, \quad TS_\tau T = S_{-\tau}. \quad (6.4)$$

For example, the last identity follows from $f(-(-t + \tau)) = f(t - \tau)$. A shift is invertible but $S_\tau^2 = S_{2\tau}$ is not generally the identity. The July source's $D(t) = Q(t + \pi/(2\omega))$ therefore relates complete signals; it does not by itself specify an involution or a causal update on integer passes [P06, physical pp. 3,8, Eq. (1) and equation-index entry (1)]. For a real monochromatic sinusoid and nonzero ω , quadrature gives zero inner product when integrated over a full period. The constant-signal counterexample shows why that orthogonality is not a general property of the displayed time shift. Its use in a sampled recursion also requires interpolation, continuation and clock conventions.

A spatial mirror across a plane through $\mathbf{x}_0$, with specified unit normal $\mathbf{n}$, is instead

$$\mathcal{P}\mathbf{x} = \mathbf{x} - 2\mathbf{n}[\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0)]. \quad (6.5)$$

Decomposing $\mathbf{x} - \mathbf{x}_0$ into tangential and normal components proves $\mathcal{P}^2 = I$. The later geometric manuscript also writes $Q^* = (g(C), H)$, but does not specify the component functions of g [P14, physical p. 10, §4.2]. Neither the star notation nor a reflected tetrahedron establishes complex conjugation or an action on the original full glyph state.

For an autonomous bijective map F , a precise reversal condition is an involution $\mathcal{T}$ satisfying

$$\mathcal{T}F\mathcal{T} = F^{-1}. \quad (6.6)$$

This is a definition of reversibility, not a property asserted of the incomplete source recursion. A nonautonomous map additionally needs the clock and prescribed forcing reversed; a stochastic model needs a specified transformation of its noise. Oddness, reciprocity, invertibility or a spatial reflection alone does not prove (6.6).

6.3 The later real coupled-field formulation

The March 2026 program initializes two independent real arrays R, D , with the second draw scaled by 0.01 [Infinity, lines 70–91]. Its historical labels do not impose $D = iR$, $D = -R$, or the July time-shift constraint. These arrays constitute a different state carrier from the June constrained complex pair.

Freeze the current geometry and let W be its inverse-distance, row-normalized neighbor matrix. On nonempty rows the simultaneous amplitude substage is

$$\begin{pmatrix} R' \\ D' \end{pmatrix} = B_p \begin{pmatrix} R \\ D \end{pmatrix}, \quad B_p = \begin{pmatrix} A_R & K_p \\ K_p & A_D \end{pmatrix}, \quad (6.7)$$

where

$$A_R = 0.85I + 0.15W, \quad A_D = 0.925I + 0.075W, \quad c_p = 0.0008699 \sin(0.244p + \pi/2). \quad (6.8)$$

Here $K_p = c_p I$ when every row has neighbors. To include empty rows, set their rows of W to identity rows and the corresponding diagonal entries of K_p to zero: the program keeps both amplitudes and skips cross terms there [Infinity, lines 114–135].

Proposition 6.2 (Exchange and quarter-turn conditions). *For the fixed-geometry linear substage, sector exchange $S(R, D) = (D, R)$ commutes with B_p exactly when $A_R = A_D$. The real quarter-turn $J(R, D) = (-D, R)$ commutes exactly when $A_R = A_D$ and $K_p = 0$.*

Proof. Block multiplication gives

$$SB_p S = \begin{pmatrix} A_D & K_p \\ K_p & A_R \end{pmatrix}.$$

Equality to B_p is equivalent to equality of the diagonal blocks. Likewise, comparing

$$JB_p = \begin{pmatrix} -K_p & -A_D \\ A_R & K_p \end{pmatrix}, \quad B_p J = \begin{pmatrix} K_p & -A_R \\ A_D & -K_p \end{pmatrix}$$

gives both asserted conditions. $\square$

For the printed coefficients, equality requires $0.075(W - I) = 0$ as an operator on the chosen domain. On a restricted invariant subspace it may hold, for example on spatially constant arrays when $W\mathbf{1} = \mathbf{1}$. It does not hold for general nonuniform inputs.

Subsequent program stages use $H(p)$ and $H(p + \pi/2)$ for the two amplitudes, position fusion weighted by R only, and a sine multiplier on R only, followed by the signed compressor and position noise [Infinity, lines 162–202]. Consequently, conditional exchange in one substage does not establish exchange symmetry of the whole update. The cross term in (6.8) uses $0.244p$, while the later multiplier uses $2\pi 0.244p$; these are different numerical frequencies.

For fixed geometry and fixed clock, the amplitude-only map is odd: its precompression stages are linear and C_s is odd. Simultaneous sign inversion is therefore an equivariance of that conditional map. It fails for the full position/amplitude update: the center is weighted by $|R|$, but the fusion displacement is multiplied by signed R . Holding positions fixed while reversing amplitudes changes that displacement.

There is a different supported covariance. Negate all positions, retain R, D , and negate the position-noise realization. Distances and graph weights stay fixed; componentwise tanh, centers and displacements transform with the position sign. The full formula therefore transforms by spatial inversion. A symmetric Gaussian noise law gives invariance in distribution; using the same nonzero noise realization does not give pathwise equality. General rotations do not follow, because componentwise tanh is not equivariant under arbitrary rotations.

Nor does the coupling preserve the whole nonnegative cone. If $c_p < 0$, choose a nonempty row with $R = 0$, zero neighboring R and $D > 0$. Its new R is negative. This is an algebraic counterexample for arbitrary inputs, not a claim about the signs of previously accepted trajectories. The later program thus supports particular conditional symmetries, not a general identification of the two sectors.

6.4 The June tensor expression remains separately typed

The collected June model uses

$$\kappa(T) = \nabla(\nabla \cdot T) - \Delta T. \quad (6.9)$$

For a smooth three-dimensional vector field with Euclidean derivatives, this equals $\nabla \times (\nabla \times T)$ by the standard component identity. For a scalar in several dimensions the divergence is not defined by those conventions; higher tensors require specified contractions. The source does not thereby determine a unique tensor type [P13, physical pp. 3,5,8–9].

For spatially constant phase, linearity gives $\kappa(e^{i\theta}T) = e^{i\theta}\kappa(T)$. A spatially varying phase introduces product-rule terms. The master expression on physical p. 5 places generally complex curvature inside a floor while projecting only a separate coupling term to its real part. There is no canonical ordered-real floor on complex numbers. A real projection or componentwise convention would be additional structure, not an automatic consequence of the quarter-turn relation.

The same source supplies both a prescribed phase schedule $\theta_n = n\pi/(2N_{\text{iter}})$ and a feedback rule

$$\theta_{n+1} = \theta_n + \epsilon_\theta \text{ mean } |Q_{\text{RDPTF}}|. \quad (6.10)$$

We use ϵ_θ to distinguish this coefficient from a logarithmic regularizer. No rule chooses one schedule as an approximation to the other [P13, physical pp. 8–9]. For $\epsilon_\theta \geq 0$ the unwrapped phase is nondecreasing; constant nonzero input magnitude makes it grow linearly. Reducing phase modulo 2π bounds a coordinate, not the tensor amplitude. These observations retain a well-defined phase identity while leaving the larger recursion’s typing and composition open.

7 Memory and stability statements with explicit domains

7.1 A prescribed schedule and a stored trail

Persistent IDs and recorded trails supply one meaning of memory [P01, physical pp. 2,6] [P03, physical p. 2]. The numerical REFU expression supplies another:

$$H(t) = 2e^{-t/2} \cos t + e^{-0.3t} \sin t + 1.5e^{-0.7t} + e^{-0.4t}. \quad (7.1)$$

It depends on the clock and fixed coefficients, with no glyph-history input [P07, physical p. 6, §3.1] [REFU, lines 5–13]. Calling it memory does not turn it into an accumulated field record. The associated factor $1 + \lambda H / \max(H)$ needs a normalization domain and stage placement, which the incomplete composition does not settle [P07, physical pp. 5–6].

For $t \geq 0$,

$$|H(t)| \leq 2e^{-t/2} + e^{-0.3t} + 1.5e^{-0.7t} + e^{-0.4t} \longrightarrow 0. \quad (7.2)$$

The triangle inequality proves the bound. The schedule is not globally nonnegative: at $t_n = 3\pi/2 + 2\pi n$,

$$e^{0.3t_n} H(t_n) = -1 + 1.5e^{-0.4t_n} + e^{-0.1t_n} \longrightarrow -1. \quad (7.3)$$

At integer multiples of 2π all its nonzero terms are positive. Thus the clock function has positive and negative values arbitrarily late, even though both are small. These continuous-clock statements do not assert signs on every particular discrete sampling schedule.

The later program’s shifted factor $H(t + \pi/2)$ changes the decay coefficients as well as the trigonometric phases: a term e^{-at} acquires the factor $e^{-a\pi/2}$. Different decay rates receive different factors and the nonoscillating terms remain. It is not a pure quadrature pair merely because the clock shift is $\pi/2$ [Infinity, lines 176–182].

7.2 Weighted embeddings are observers

P01’s weights and embedding can be written, with S_j denoting the summed normalized amplitude of trail j , as

$$S_j = \sum_{t \in j} R_{\text{norm},t}, \quad w_j = \frac{S_j}{\sum_k S_k}, \quad E = \sum_j w_j v_j, \quad (7.4)$$

where v_j represents trail information [P01, physical p. 7, §3.5]. The denominator must be nonzero. If every $S_j \geq 0$, these are convex weights. Likewise, nonnegative R with a positive maximum ensures $R_{\text{norm}} = R / \max R \in [0, 1]$; unrestricted signed inputs do not guarantee this range. These are sufficient sign conditions, not necessary ones: uniformly negative sums can also normalize to positive weights.

Summation is not injective. With equal weights, the pairs $(-1, 1)$ and $(0, 0)$ give the same scalar embedding, and reordering samples does not change a sum. An embedding can be useful while discarding information; lossless retention of all ancestral memory does not follow. The surviving echo observer likewise measures distances between different positions along one selected glyph’s trail, then min–max normalizes the distance matrix. It is a readout of one trajectory, not a two-glyph feedback rule; its normalization also needs nonzero range [Echo, lines 10–43].

7.3 A later state-driven memory equation

The November geometric model explicitly introduces history through

$$\dot{H}(t) = \eta I(t) - \kappa H(t), \quad I(t) = \int_{\Omega} u(\mathbf{r}, t) w(\mathbf{r}) d^3r \quad (7.5)$$

[P14, physical pp. 5,17, Eqs. (10),(25)]. Suppose I is locally integrable, η, κ are fixed finite coefficients and $H(0)$ is specified. Multiplying by $e^{\kappa t}$ and integrating yields the unique absolutely continuous solution

$$H(t) = e^{-\kappa t} H(0) + \eta \int_0^t e^{-\kappa(t-s)} I(s) ds. \quad (7.6)$$

Uniqueness follows because the difference of two solutions solves $\dot{h} = -\kappa h$, $h(0) = 0$. This is a precise field-history meaning of memory, conditional on the specified input u . It does not solve the coupled field equation or identify this later H with July's (7.1).

The same source states a wrapped drift–diffusion balance. Integrating its divergence terms gives zero when the boundary fluxes cancel; total mass is then conserved if the remaining source has zero spatial integral [P14, physical p. 17, Eq. (24)]. Boundary cancellation and zero source integral are hypotheses, not consequences of the word recursion.

7.4 Positive local results and their limits

One June scalar module has a complete standard stability argument:

$$G(x) = \frac{\phi}{\sqrt{3}} \tanh x, \quad x \in \mathbb{R}. \quad (7.7)$$

This is the printed $\phi\sqrt{3}/3$ coefficient [P08, physical p. 1, §2.1]. Since $|G'(x)| \leq \phi/\sqrt{3} < 1$, the mean-value theorem makes G a global contraction on the complete real line. Its fixed point is uniquely zero, which is directly a fixed point. The standard contraction principle concerns this map, not every composition in the collected model.

A separate drift increment is $\Delta\Theta_i = \alpha(\Theta_{\text{ref}} - \Theta_i)$ [P03, physical p. 13, §5.4]. If this is added to a numeric state with a fixed reference and fixed $0 < \alpha < 2$, the new error is $(1 - \alpha)$ times the old error, a strict contraction in any norm. The source does not specify the reference rule, bounds on its memory-modulated coefficient, or placement relative to the implicit update. This conditional result is useful without electing a complete dynamics.

7.5 A literal series and a geometric contraction claim

The June collected manuscript prints

$$\text{Revolution}(X) = \sum_{n=1}^{\infty} \Omega_X \left(1 + \frac{\beta_n}{\alpha_n} \Omega_X \right) \quad (7.8)$$

[P13, physical pp. 4,6]. For a fixed finite nonzero Ω_X , every term needs $\alpha_n \neq 0$. Convergence requires its terms to vanish, hence $\beta_n/\alpha_n \rightarrow -1/\Omega_X$. This necessary condition is not sufficient: writing $\beta_n/\alpha_n = -1/\Omega_X + a_n$ reduces the series to $\Omega_X^2 \sum_n a_n$, which diverges for $a_n = 1/n$. If the ratio tends to zero instead, the terms tend to $\Omega_X \neq 0$. The prose description of a self-modifying constant does not supply an indexed update for Ω_X or a convergence prescription. Treating it as Ω_n would create an additional definition.

The November field pullback takes the form

$$u^+ = u \circ \Phi, \quad \Phi(\mathbf{r}) = \mathbf{r} - \lambda_c \mathbf{n}(\mathbf{r}), \quad \lambda_c > 0. \quad (7.9)$$

Under its stated assumption that Φ maps Ω onto itself,

$$\|u \circ \Phi - v \circ \Phi\|_\infty = \|u - v\|_\infty.$$

This is norm preservation, not strict contraction of fields. An L^2 change of variables instead involves the Jacobian [P14, physical p. 16, Eqs. (17)–(18)].

There is a separate obstruction to the printed position-space argument. For a differentiable unit normal, $\mathbf{n}^T D\mathbf{n} = 0$, obtained by differentiating $\mathbf{n}^T \mathbf{n} = 1$. Therefore $D\mathbf{n}$ is singular and $D\Phi = I - \lambda_c D\mathbf{n}$ has eigenvalue 1. Its induced operator norm cannot be strictly less than one. A fixed point of the printed Euclidean-coordinate map would moreover require $\mathbf{n} = 0$, inconsistent with a unit normal and $\lambda_c > 0$. A vector field vanishing at the target might avoid this obstruction, but is not the given map.

Even in a different map, eigenvalues inside the unit circle do not alone imply a contraction in an already specified norm for a nonnormal matrix. And a contraction on a complete invariant space has one fixed point, not a nontrivial cycle: if $F^m x = x$ and its contraction factor is $\rho < 1$, then $d(x, Fx) \leq \rho^m d(x, Fx)$, so $Fx = x$. Thus the source’s accompanying alternative of a small set of limit cycles does not follow from its contraction argument [P14, physical pp. 16–18]. The exact memory solution and conditional mass balance survive these limitations; a complete attractor theory still needs a defined field map, domain, boundary conditions and proof.

8 Later geometric and finite-core comparisons

Later sources can clarify the range of mathematical meanings attached to recurring names without completing the earlier model retrospectively. The November geometric manuscript’s elementary identity is

$$|\sin(3\theta + \delta) + \sin(3\theta - \delta)| = 2|\sin(3\theta) \cos \delta|. \quad (8.1)$$

For $\cos \delta \neq 0$ the magnitude has period $\pi/3$; for $\delta = \pi/2$ modulo π it vanishes [P14, physical p. 11, §4.4]. This establishes a sixfold magnitude pattern, not a sixth-harmonic field or the dynamical emergence of tetrahedra.

Reflection also does not force a Star-of-David projection. Take the regular tetrahedron with base

$$(\cos(2\pi k/3), \sin(2\pi k/3), -1/\sqrt{8}), \quad k = 0, 1, 2,$$

and apex $(0, 0, 3/\sqrt{8})$. All six edges have length $\sqrt{3}$. Reflecting in $z = 0$ reverses its orientation but gives the same XY projection: a triangle and its center, not two rotated triangles. This counterexample limits a universal reading of the mirror-projection claim without ruling out a differently specified geometry [P14, physical pp. 10–11].

The other November manuscript defines a six-component complex model in the ordered basis $(0+, 1+, 2+, 0-, 1-, 2-)$ [P15, physical pp. 3–6]. Following its stated per-corridor helicity mixing, the order is sector-major and its four factors reduce to

$$U = A \otimes B, \quad A = e^{-i\alpha\sigma_z} e^{-i\beta\sigma_x}, \quad B = MDC. \quad (8.2)$$

Appendix B specifies the factors and the basis qualification. A product operator cannot create corridor/helicity entanglement from a product input. A formula written for a differently ordered basis must permute all operators consistently before it can be compared.

The same appendix proves the conditional norm identity, parameter conjugacy and generator-sign comparisons. In particular, explicit normalization in the source explains normalized reported vectors without making the underlying linear map unitary [P15, physical pp. 24,44–45].

The source's identification of three spin-half tensor factors with six dimensions is a separate representation issue: the tensor product has dimension eight and no total magnetic weight zero. A six-dimensional direct sum is a different construction [P15, physical p. 49, Appendix O].

These later comparisons are deliberately limited. They do not certify all later spectral tables, reopen the completed Tower II experiments, or change the corrected basis and factor-order conventions used by the accepted QSM/Bridge audits. In particular, a conditional factorization of one stated per-corridor action is not a claim that every later implementation uses that action.

9 Definition requirements and conclusions

9.1 What the reconstruction establishes

The historical materials determine several useful structures. The glyph schema identifies the intended ingredients of a state. The printed feedback, compression maps and observer formulas can be compared once their domains are explicit. The exact branch theorem determines scalar existence and uniqueness for the implicit temporal relation without changing its coefficient. The June quarter-turn, July shift, geometric reflection and later sector matrices support distinct, precisely conditional identities. The later driven memory equation has an explicit history representation.

These results support a mathematical reconstruction rather than a selected replacement dynamics. In particular, Theorem 3.1 concerns fixed-data scalar solvability, not complete well-posedness of the glyph system. On its unique positive-threshold branch the root is a linear function of b , but that local formula does not supply roots in the missing region, choose a member of the $b = 0$ continuum, or define updates for the rest of the tuple.

9.2 The definitions still needed for a complete dynamics

A future complete model would have to specify the following mutually related items. This is a definition checklist, not an implementation plan.

1. A numeric state carrier, its constraints and norm, and the distinction between evolving state, persistent labels and readouts.
2. Required time levels and initialization, including whether the temporal difference generates motion, constrains it, or observes it.
3. A spatial metric, dimensional scales, boundary conventions and any graph construction or refresh rule.
4. Phase, memory and echo-set laws, including exceptional cases such as stationary trails, empty neighborhoods and zero normalization denominators.
5. A selected compressor and its stage order, with sign, zero, threshold, and any partner-timing conventions.
6. A fusion operation if fusion changes state, rather than merely a predicate applied to observations.
7. For an implicit equation, an admissible domain with existence and uniqueness conditions, or an explicit multivalued/selection interpretation.
8. A stated symmetry or reversor acting on every relevant state, parameter, clock and noise component.

None of these can be replaced by the recurrence of an operator name. Nor can the later source's real R/D arrays supply the missing typing of the June tensor or July tuple by assumption.

9.3 Completion comparisons are not recovered intent

There are several mathematically distinct future choices. One could formalize the surviving explicit formula (2.2); one could introduce a new implicit model with a globally Lipschitz compressor; or one could separate temporal curvature from the update and supply a velocity state. Each needs a declaration and subsequent analysis. None is selected here.

For example, the standard sufficient condition for a newly specified equation

$$y = \tilde{C}(\chi y + d), \quad |\chi| \text{Lip}(\tilde{C}) < 1, \quad (9.1)$$

on a complete invariant domain ensures a unique fixed point. Appendix A.5 supplies the short proof and the domain qualification. The original threshold compressor does not satisfy that global Lipschitz hypothesis. Even introducing $|\chi| < 1$ alone leaves no-root examples because its jump persists. This comparison establishes what one possible completion would need; it does not approve that completion.

9.4 Scope of the manuscript

The analysis is bounded by the retained manuscripts and operative code. Source equations and counterexamples establish the claims stated here; they do not authenticate every historical experiment or infer the author's missing intent. Earlier finite arithmetic checks are reused as supporting examples, without suite replay. No field update or new trajectory was computed for the manuscript.

Physical particle labels, horizon narratives and the constant lock are historical motivations. They need independently defined observables, units and evidence before they can serve as physical conclusions. The distinct completed Tower II investigation concerns a specified later program and remains outside this paper's numerical scope.

The conclusion is therefore constructive but limited: the original mathematics already supports exact source-specific statements, including the scalar branch theorem and several conditional symmetry and memory identities. It does not determine a unique combined evolution. Preserving that boundary makes those statements available for research without silently replacing the model whose foundations are being reconstructed.

A Complete scalar branches for alternative operators

This appendix compares source-defined operators under substitution into $y = C(y + b)$. It does not assert that the original paper requested those substitutions. All variables are finite real numbers unless stated otherwise. To avoid a collision of roles, the plateau values are denoted $a_- = e^{-L}$ and $a_+ = e^L$, a fixed partner cap is m , and a newly introduced temporal coefficient is χ . These are presentational renamings, not changes in the source equations.

A.1 Fixed partner cap

With the partner held fixed, P03's minimum map gives $y = \min(y + b, m)$ [P03, physical p. 6, §2.5]. Every root satisfies $y \leq m$. If $y < m$, equality requires $b = 0$; if $y = m$, it requires $b \geq 0$.

Consequently the complete set is

$$\mathcal{S}_{\min}(b; m) = \begin{cases} \emptyset, & b < 0, \\ (-\infty, m], & b = 0, \\ \{m\}, & b > 0. \end{cases} \quad (\text{A.1})$$

Each listed candidate satisfies the original minimum expression, proving sufficiency. Nonnegative amplitudes require intersection with $[0, \infty)$; a negative cap then admits no root. If the partner is an unknown successor, its coupling must be analyzed separately and this fixed-cap classification is insufficient.

A.2 Positive-output logarithmic map

Let $L > 0$, $\epsilon > 0$, $x = y + b$, and use (4.4). Every root is positive. The following exhaustive table includes clamp boundaries in adjacent branches intentionally; duplicate candidates are removed from the union.

Input branch	Candidate or family	Necessary and sufficient tests
$x = 0$	$y = 1$	$b = -1$
$x > 0$, lower	$y = a_-$	$a_- + b > 0$ and $a_- + b + \epsilon \leq a_-$
$x > 0$, interior	Eligible y , only if $b = -\epsilon$	$y - \epsilon > 0$ and $a_- \leq y \leq a_+$
$x > 0$, upper	$y = a_+$	$a_+ + b > 0$ and $a_+ + b + \epsilon \geq a_+$
$x < 0$, lower	$y = a_-$	$a_- + b < 0$ and $(\epsilon - a_- - b)^{-1} \leq a_-$
$x < 0$, interior	$y_{\pm}$ in (A.2)	Real discriminant; $y_{\pm} > 0$; $y_{\pm} + b < 0$; $a_- \leq y_{\pm} \leq a_+$
$x < 0$, upper	$y = a_+$	$a_+ + b < 0$ and $(\epsilon - a_+ - b)^{-1} \geq a_+$

On the negative-input interior, equality is

$$y = \frac{1}{\epsilon - y - b} \iff y^2 + (b - \epsilon)y + 1 = 0,$$

so the candidates are

$$y_{\pm} = \frac{\epsilon - b \pm \sqrt{(b - \epsilon)^2 - 4}}{2}. \quad (\text{A.2})$$

The table's sign and clamp conditions exclude extraneous polynomial roots. On the positive-input interior, equality is $y = y + b + \epsilon$, requiring $b = -\epsilon$. On each plateau the only possible output is its endpoint. Zero input has the separate value one. These cases exhaust all real inputs and prove the table.

At the historical values $L = 10$, $\epsilon = 10^{-6}$ and offset $b = -3$, both roots of (A.2), approximately 0.381966 and 2.618035, have negative input and lie between the plateaus. Both solve the substituted relation. At $b = -\epsilon$, every eligible positive-input interior value is a root. Thus this alternate compressor does not generally restore unique solvability. If $L = 0$ instead, the map is constant one and $y = 1$ is the unique root for every offset; that degenerate case is outside the table's assumption.

A.3 Signed magnitude clamp

For (4.5), the complete candidate union is:

1. $y = 0$, admissible exactly when $b = 0$.
2. For each $s \in \{-1, +1\}$, the lower plateau $y = sa_-$, admissible exactly when $s(y + b) > 0$ and $|y + b| + \epsilon \leq a_-$.
3. For each s , the upper plateau $y = sa_+$, admissible exactly when $s(y + b) > 0$ and $|y + b| + \epsilon \geq a_+$.
4. On the interior with sign s , all y satisfying

$$b = -s\epsilon, \quad sy > 0, \quad s(y + b) > 0, \quad a_- \leq |y| \leq a_+.$$

To prove completeness, separate zero and the two input signs. On a sign- s interior, the equation is $y = y + b + s\epsilon$; on a plateau the output is its signed endpoint. The listed conditions are exactly those branch equations and their domains.

For the later defaults $a_- > \epsilon > 0$ and $b = 0$, neither interior is possible. The lower plateau would require $a_- + \epsilon \leq a_-$ and fails. Both upper plateaus succeed, as does zero. There are exactly three roots, $0, \pm a_+$.

A.4 A newly introduced temporal coefficient

If, as a new comparison only, a coefficient χ multiplies the centered temporal term, known current x , previous z and feedback F give

$$y = C_\lambda(\chi y + d), \quad d = (1 - 2\chi)x + \chi z + F. \quad (\text{A.3})$$

The full branch classification, including singular coefficients, is:

- If $\chi \neq 1$, the lower candidate is $d/(1 - \chi)$, valid exactly when that value is less than λ . If $\chi = 1$, the lower branch exists exactly for $d = 0$ and then contains all $y < \lambda$.
- If $\chi \neq 1/q$, the upper candidate is $qd/(1 - q\chi)$, valid exactly when $d/(1 - q\chi) \geq \lambda$. If $\chi = 1/q$, the upper branch exists exactly for $d = 0$ and then contains all $y \geq q\lambda$.

Indeed the two branch equations are $y = \chi y + d$ and $y = q(\chi y + d)$. Solving them and retaining their input inequalities, with zero denominators handled separately, proves necessity and sufficiency.

For $\chi = 1/2, \lambda = 1, d = 0.4$, only the lower candidate $y = 0.8$ is admissible; the upper candidate has input below the threshold. For $d = 0.6$, the lower candidate 1.2 is inadmissible and the upper candidate's input is approximately $0.8683 < 1$, also inadmissible. There is no root despite $|\chi| < 1$. The threshold jump, rather than merely the coefficient of the implicit term, is consequential.

A.5 A standard sufficient theorem for a new model

Let E be a Banach space and let $\tilde{C} : E \rightarrow E$ be globally Lipschitz with constant K . For fixed $d \in E$, define $T(y) = \tilde{C}(\chi y + d)$ and suppose $\rho = |\chi|K < 1$. Then

$$\|T(y) - T(z)\| \leq \rho \|y - z\|.$$

The successive differences of the abstract iteration $y_{m+1} = T(y_m)$ are at most $\rho^m \|y_1 - y_0\|$. Their geometric-series bound makes (y_m) Cauchy; completeness gives a limit y_* and continuity gives $T(y_*) = y_*$. Two fixed points satisfy $\|y_* - z_*\| \leq \rho \|y_* - z_*\|$ and hence coincide. This is the standard contraction proof, included to state all needed assumptions. The same argument works

on a closed invariant subset. Without invariance a local Lipschitz estimate does not establish existence in the intended domain.

This is a sufficient comparison theorem for a newly specified compressor and state carrier. It is neither an adopted SRG law nor an executed iteration. It cannot be applied to the discontinuous threshold map by omitting that map's branch conditions.

B The later finite-core carrier and operator conventions

B.1 Basis, action and factorization

This appendix is conditional on the per-corridor helicity action stated in the November manuscript. It is not a re-audit of all its tables or of any corrected kernel. Write the ordered basis as

$$(|0, +\rangle, |1, +\rangle, |2, +\rangle, |0, -\rangle, |1, -\rangle, |2, -\rangle).$$

For matrix construction this order identifies the carrier with $\mathbb{C}_{\text{sector}}^2 \otimes \mathbb{C}_{\text{corridor}}^3$, with the sector index outermost. Let

$$s = \frac{1}{\sqrt{3}}(1, 1, 1)^T, \quad P = ss^\dagger, \quad C = (1 - \lambda_c)I_3 + \lambda_c P, \quad (\text{B.1})$$

$$M = e^\gamma P + e^{-\eta}(I_3 - P), \quad D = \text{diag}(1, e^{-i\varphi}, e^{-2i\varphi}). \quad (\text{B.2})$$

The full projector is $I_2 \otimes P$. In this ordering the four per-corridor factors are

$$\begin{aligned} U_{\text{RPCO}} &= I_2 \otimes C, & U_{\text{flip}} &= e^{-i\beta\sigma_x} \otimes I_3, \\ U_{\text{TGMO}} &= e^{-i\alpha\sigma_z} \otimes D, & U_{\text{REFU}} &= I_2 \otimes M. \end{aligned} \quad (\text{B.3})$$

These express the literal projector, phase and amplification definitions [P15, physical pp. 3–6, §§2–3]. With the printed factor order $U = U_{\text{REFU}}U_{\text{TGMO}}U_{\text{flip}}U_{\text{RPCO}}$, the elementary identity $(A \otimes B)(C \otimes D) = AC \otimes BD$ proves (8.2). It follows directly that $U(h \otimes v) = Ah \otimes Bv$, so product inputs remain products.

The explicit index mapping matters:

Convention	Ordered labels	Same-corridor pairs
Printed sector-major	0+, 1+, 2+, 0–, 1–, 2–	(0, 3), (1, 4), (2, 5)
Corridor-major	0+, 0–, 1+, 1–, 2+, 2–	(0, 1), (2, 3), (4, 5)

Appendix C writes $I_3 \otimes e^{-i\beta\sigma_x}$ beside the sector-major diagonals [P15, physical p. 24]. That Kronecker product mixes adjacent entries, hence implements the stated per-corridor action in corridor-major order, not the printed order. The two complete representations are related by a permutation matrix Π , with every operator transformed as $\Pi U \Pi^{-1}$. Changing the flip alone while retaining the other diagonals is not such a basis change. The factorization above follows the stated action and declared order; it does not silently certify the inconsistent literal combination.

B.2 Norm, normalization and sector conjugacy

At $\varphi = 2\pi/3$, $s^\dagger D s = (1 + e^{-2\pi i/3} + e^{-4\pi i/3})/3 = 0$, while $Cs = s$. For real α, β , A is unitary. Therefore

$$\|U(h \otimes s)\|^2 = e^{-2\eta} \|h\|^2. \quad (\text{B.4})$$

At the source value $\eta = 0.423$, the factor is approximately 0.4291. The raw map does not preserve norm. The source explicitly renormalizes after each update in Appendix C and again in its implementation discussion [P15, physical pp. 24,44–45]. The map $\psi \mapsto U\psi/\|U\psi\|$ is a nonlinear normalized map defined where $U\psi \neq 0$, rather than a linear unitary operator.

For exchange $S = \sigma_x \otimes I_3$, use $\sigma_x \sigma_z \sigma_x = -\sigma_z$ and $\sigma_x^2 = I$ to obtain

$$SU(\alpha, \beta)S = U(-\alpha, \beta). \quad (\text{B.5})$$

This is conjugacy between parameter values, not exchange invariance at the source's fixed $\alpha = 2\pi(0.244)$. Also $e^{-i\beta\sigma_x} = \cos \beta I - i \sin \beta \sigma_x$ is a complete swap up to a global phase only when $\beta = \pi/2$ modulo π . At $\beta = \pi(0.244)$ it is partial mixing [P15, physical pp. 5–6].

For $0 \leq \lambda_c < 1$, the eigenvalues of C are 1 and $1 - \lambda_c$, so all factors are invertible; at $\lambda_c = 1$, C is singular. Even in the invertible case, reversing the factor order and inverting amplification/damping does not establish a physical reversor of the form (6.6).

B.3 Generator convention and finite-step limitations

Under the convention $U = e^{-iH}$, choosing the memory generator continuously from zero strength gives

$$H_{\text{mem}} = i[\gamma P_{\text{core}} - \eta(I - P_{\text{core}})], \quad e^{-iH_{\text{mem}}} = e^{\gamma P_{\text{core}} - \eta(I - P_{\text{core}})}. \quad (\text{B.6})$$

The opposite sign printed on physical pp. 8 and 31 instead inverts the stated amplification and damping [P15, Appendix E.5]. Other logarithm branches require separate choices; the exponential alone does not give a unique generator.

If $\lambda_c = \kappa_c \Delta t$, expansion of $I + \kappa_c \Delta t(P_{\text{core}} - I)$ gives $H_{\text{comp}} = i\kappa_c(P_{\text{core}} - I)$ to first order under the same convention. This differs both in sign and in its identity term from the printed $-i\kappa_c(P_{\text{core}} - I/3)$ [P15, physical pp. 7,31]. For the complete finite operator, $H_{\text{eff}} = i \log U$ with a specified branch; the pseudocode's $-i \log U$ has the opposite sign [P15, physical p. 34, Appendix H.6]. Noncommuting finite factors do not generally have a generator equal to the sum of their separate generators. A small-step approximation needs scaled strengths and control of the commutator terms.

B.4 The representation claim

The later spin-network identification asserts a tensor product of three spin-half edge spaces while counting three edges times two helicities as six states [P15, physical p. 49, Appendix O]. A tensor product has dimension $2^3 = 8$, whereas three doublets in direct sum have dimension six. In the tensor product the total magnetic weights are $\pm 3/2$ and $\pm 1/2$; none is zero. An invariant singlet would be annihilated by total J_z , so no such singlet exists here. This elementary obstruction concerns that identification, not the existence of the separate six-component carrier or all possible representations that one might subsequently define.

C Source dictionary, parameter roles and physical interpretations

The references below give complete source titles and date evidence. All P-number page locators refer to physical PDF pages. Code locators refer to retained original lines; the review ledger disambiguates paths, records hashes, and connects manuscript claims to the page/line evidence. An archive ID is not a DOI, and no local path is offered as a public link.

C.1 Several state carriers

Source and date	Carrier actually stated	Boundary of the identification
June P09–P13	Phase-labelled tensor/field expressions	Tensor type and composite floor need definition
July P03	Glyph tuple and symbolic trail state	Tuple schema is not a scalar algebra
July P05/code	Positions on a fixed neighbor graph	No full resonance/echo-state update
July P06	Two signals related by a time shift	Not arbitrary independent sectors
November P14	Field, geometry and driven memory	Domains and nonlinear field terms remain required
November P15	Six complex components	Basis action and factor order must be fixed
March Infinity	Two real amplitude arrays and positions	Not constrained by the June complex relation

C.2 Constants require a role and an independent estimator

The fusion and dark-sector manuscripts print

$$\pi e\phi = \gamma^{-1}\zeta(3) \iff \gamma = \frac{\zeta(3)}{\pi e\phi} \approx 0.0869947475 \quad (\text{C.1})$$

[P01, physical p. 3, Eq. (1)] [P06, physical p. 4, Eq. (2)]. The latter’s symbol appendix also identifies gamma as the Euler–Mascheroni constant, approximately 0.577 [P06, physical p. 6]. These are incompatible values under the same literal identity, not a numerical uncertainty. The position program’s unused GAMMA=0.577, the later finite-core reinforcement 0.577 and the Infinity coupling parameter 0.08699 play different roles. A shared letter does not derive a relation among them.

The later URF text describes a run-dependent coupling inferred from spectra, while its algorithm leaves GammaEstimate unspecified [P19, physical pp. 8–9,13]. Solving (C.1) for gamma is not an independent empirical test of that identity. An estimator would need a definition independent of the target equality and a declared observable. Similarly, the source uses of λ_{VP} as an overlap length, clustering radius or numerical 0.618 require different parameter dictionaries. The number 0.244 cannot silently be radians per pass, cycles per pass and hertz without a conversion and physical clock.

C.3 Horizon equations do not specify a reversal map

The July manuscript’s horizon expression is

$$\Psi_H(t, r) = A_H e^{-\Gamma t} \cos(\omega_r r - \varphi_0) \quad (\text{C.2})$$

[P06, physical p. 4, Eq. (3)]. At fixed r it has decay but no temporal cosine. The coefficient multiplying radius is labeled in hertz in the symbol discussion; a spatial scale and dimensional convention are missing. The expression does not supply a scattering boundary condition or a time-reversal action.

For one mode the printed threshold integral evaluates to

$$\Lambda_{VP} = \int_0^{R_c} \lambda_n e^{-\alpha_n r} dr = \begin{cases} \lambda_n (1 - e^{-\alpha_n R_c}) / \alpha_n, & \alpha_n \neq 0, \\ \lambda_n R_c, & \alpha_n = 0. \end{cases} \quad (\text{C.3})$$

No sum over n is printed, so adding one would be a further definition [P06, physical p. 4, Eq. (5)]. The stated entropy balance similarly implies only

$$\frac{d}{dt}(S_{\text{child}} + S_{\text{parent}}) = \epsilon_f. \quad (\text{C.4})$$

Conservation needs $\epsilon_f = 0$ [P06, physical p. 5, Eq. (7)]. These integral and balance identities can be retained without inferring microscopic reversed time, or identifying the historical sector labels with established particles.

D Evidence roles and assistance

The central statements are supported by the proofs and source locators in this paper. FOUNDATIONS-01 supplied twenty finite arithmetic certificates illustrating branch, sign, matrix and constant comparisons [F01, B]. They were retained as prior evidence, not rerun for a new receipt, and do not substitute for proofs of general claims. The drafting supplement supplies a claim/source ledger, a cited-page and code-line review, mathematical notes, build and page-review records, and consumed-input integrity hashes. It does not reproduce earlier trajectory archives or certify every numerical claim in the historical manuscripts.

The proposed human author is Hilmir Frímann Halldórsson. GPT supplied mathematical review of the preceding reconstruction and the opening draft, then reviewed v0.1 and requested the minor revisions E01–E03. Codex carried out source investigation, the recorded FOUNDATIONS-01 algebra checks, manuscript drafting, and the present revision with citation verification and typesetting review. The prior algebra checks were not rerun for this revision. These are assistance roles rather than coauthorship, external peer review, or publication approval. This v0.1.1 manuscript addresses E01–E03 and is returned for GPT closure review; acceptance of the revision is not claimed before that review. No affiliation, release license or publication venue is assigned.

References

- [P01] Archived manuscript P01. *Symbolic Recursion and SRG Fusion: Emergent Lattice Fields from Glyph Memory Dynamics*. Cover date: 15 July 2025. 24 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P03] Archived manuscript P03. *Symbolic Recursion and SRG Formation: Full Method and Emergence*. Cover date: 14 July 2025. 19 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P05] Archived manuscript P05. *Patch Research: Detailed Reconstruction of Missing Components in Symbolic Recursion and SRG Framework*. Cover date: 15 July 2025. 3 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P06] Archived manuscript P06. *Recursive Dark Energy and Vacuum Feedback: A Vesica-Phase Cosmological Framework*. Cover date: 13 July 2025. 9 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P07] Archived manuscript P07. *Recursive Numerical Operators for Symbolic Dynamics*. Cover date: 11 July 2025. 10 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P08] Archived manuscript P08. *Recursive Symbolic Evolution and Feedback Dynamics in Python*. Cover date: 29 June 2025. 3 physical PDF pages. Stable local archive identifier; public publication status not established.

- [P09] Archived manuscript P09. *Recursive Symbolic Evolution and Feedback Dynamics in Python*. Cover date: 29 June 2025. 7 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P10] Archived manuscript P10. *Recursive Symbolic Evolution and Its Application in Quantum and Cosmological Systems*. Cover date: 29 June 2025. 4 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P11] Archived manuscript P11. *Why the Toroidal Laplacian Failed: A Structural Explanation*. No established cover date; PDF creation metadata: 29 June 2025. 2 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P13] Archived manuscript P13. *Unified Recursive Cosmological Model Memo*. No established cover date; PDF creation metadata: 29 June 2025. 17 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P14] Archived manuscript P14. *Recursive Engines in Folded Tri-Octagon Geometry: From Symbolic Qubits to the Dual-Tetrahedral SRG Core*. Cover date: 24 November 2025. 25 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P15] Archived manuscript P15. *Recursive SRG Evolution: A Qutrit-Qubit Operator, Dual-Core Geometry, and Computational Interpretation*. Cover date: 24 November 2025. 51 physical PDF pages. Stable local archive identifier; public publication status not established.
- [P19] Archived manuscript P19. *Unified Recursive Fields: Geometry, Recursion, and Consciousness*. Cover date: 6 September 2025. 15 physical PDF pages. Stable local archive identifier; public publication status not established.
- [Infinity] Archived source code `src_infinity.py`. Retained March archive source; internal date 14 March 2026. Principal lines: 49–53,70–91,114–135,162–202. The review ledger records the exact retained path and source identity.
- [Position] Archived source code `recursive_field_evolve.py`. Original reconstruction engine; no separate composition date established. Principal lines: 31–40,45–78. The review ledger records the exact retained path and source identity.
- [RPCO] Archived source code `RPCO.py`. Original Reading material spatial helper. Principal lines: 5–13. The review ledger records the exact retained path and source identity.
- [TGMO] Archived source code `TGMO.py`. Original Reading material toroidal sampler. Principal lines: 5–14. The review ledger records the exact retained path and source identity.
- [REFU] Archived source code `REFU.py`. Original Reading material clock-function helper. Principal lines: 5–13. The review ledger records the exact retained path and source identity.
- [Attractor] Archived source code `recursive_attractor.py`. Original velocity/position program. Principal lines: 29–56. The review ledger records the exact retained path and source identity.
- [Helicity] Archived source code `helicity_REFU_TGMO_RPCO_field_evolution.py`. Original prescribed vector-field program. Principal lines: 23–40,57–63. The review ledger records the exact retained path and source identity.
- [Echo] Archived source code `echo_map_2.py`. Original `Simulations_Trash/Simulations_part_4` trail observer. Principal lines: 10–43. The review ledger records the exact retained path and source identity.
- [F01] *SRG FOUNDATIONS-01 Original Recursion and Dual-Sector Mathematics*. Codex source-reconstruction report, 10 October 2026, with subsequent GPT review. Acceptance is limited to the recorded reconstruction scope.
- [B] *Exact scalar branches and conditional completion theorem*. Codex, FOUNDATIONS-01 appendix, 10 October 2026. Analytical comparisons, not an adopted replacement model.
- [C] *Source catalog and coverage*. FOUNDATIONS-01, 10 October 2026. Titles, date evidence, identities and bounded coverage.